Flat memory for CXL GPU

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Context: GPU memory crisis

- Model size **far exceed** GPU memory capacities (VRAM)

2020

ChatGPT

GPT-3: **175B** params

350GB[1] VRAM

2025

Deepseek v3: **671B** params

~1.34 TB[1] VRAM



NVIDIA DGX B200:
1440GB VRAM

450,000€

Observation: Capacity-bound workloads, like inference, only need memory expansion — no extra compute

[1]1B params ~= 2GB

Motivation: Capacity expansion with CXL



NVIDIA DGX B200:
450,000€ – 1440GB VRAM

350€/GB



CXL memory
1000-1500€ – **128GB** DDR4

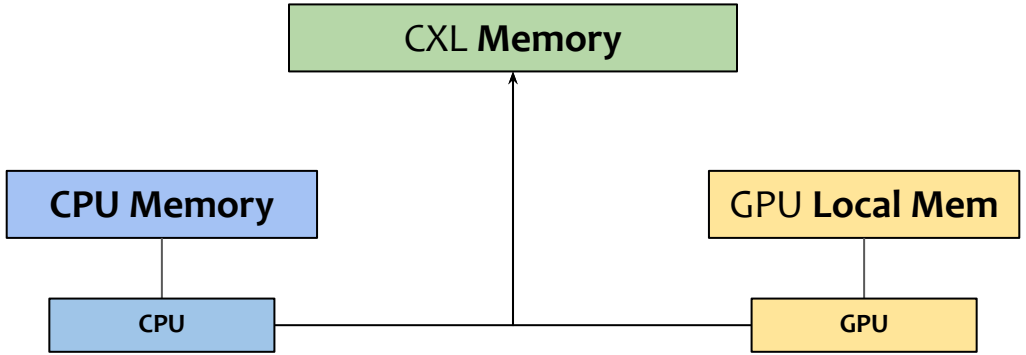
7-12€/GB

~97-98% cheaper

CXL memory decreases hardware costs up to **30 to 50** times!

CXL-enabled GPUs

- CXL memory: a separate physical memory region
- Directly addressable (`ld` / `st`)
- Cache coherent between CPU/GPU



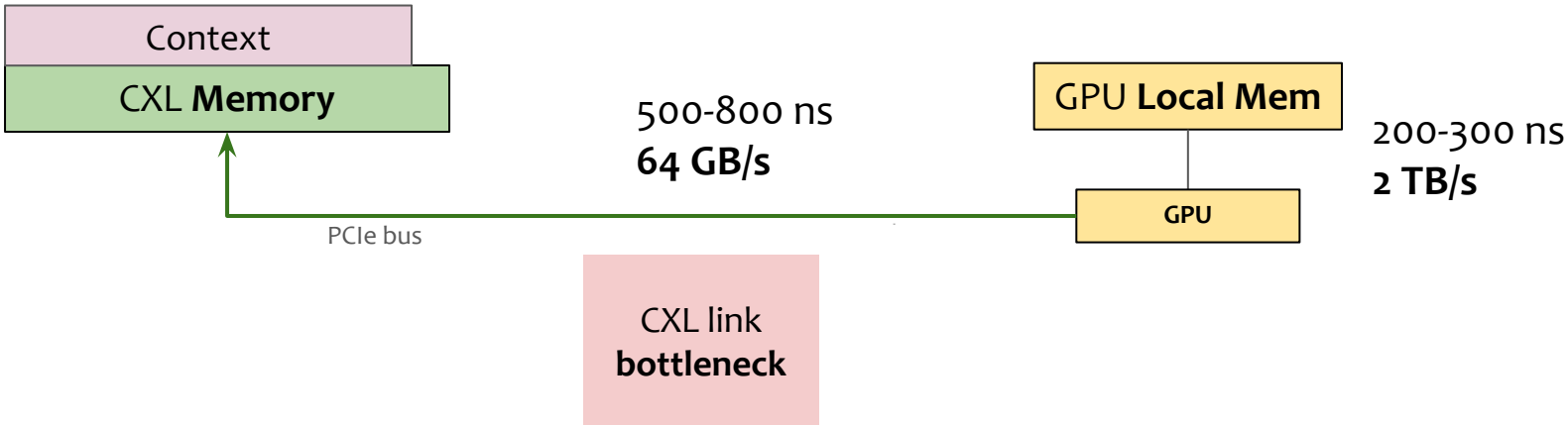
Programmers must manage **data placement**

Problem: CXL Memory is not HBM!



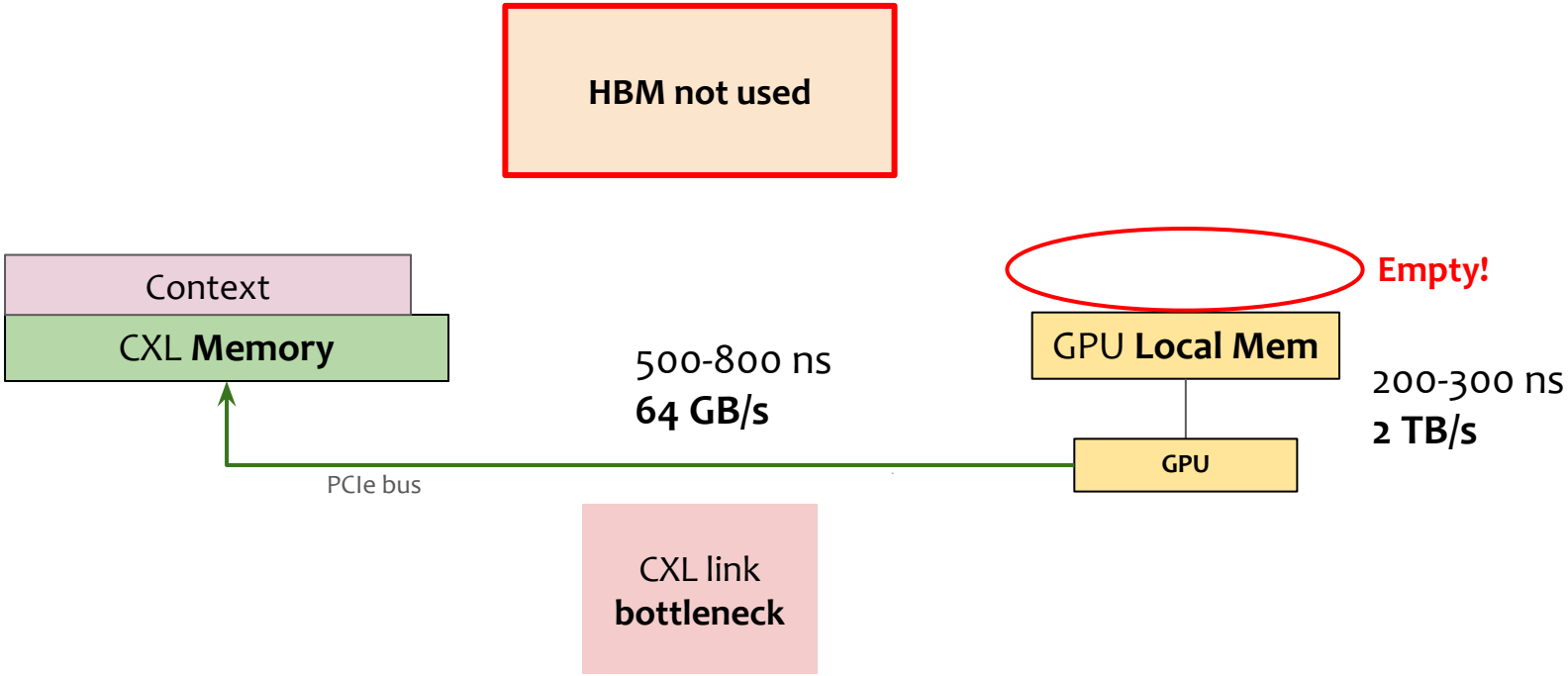
Direct memory line access

Problem: CXL Memory is not HBM!



Direct memory line access

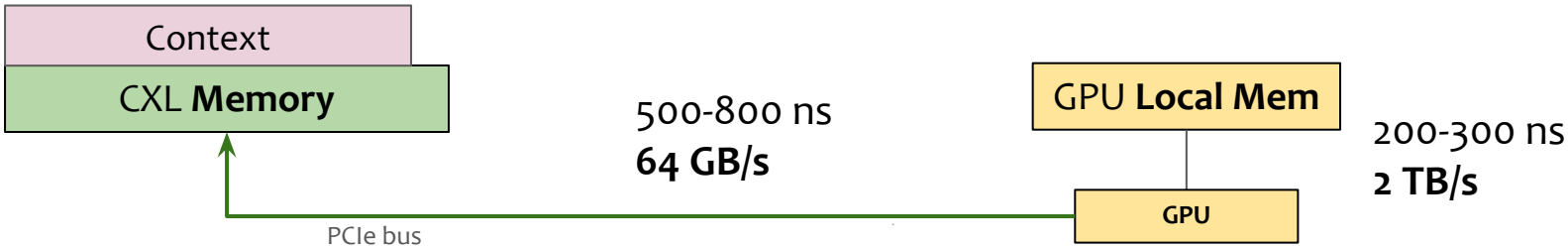
Problem: CXL Memory is not HBM!



Direct memory line access

Problem statement

Problem Statement: How to automatically hide link latency and ensure optimal CXL traffic for **small cache-line accesses?**



Direct memory line access

Proposal: Flat memory for CXL GPUs

Use GPU local HBM **as coherent cache** for CXL memory



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Direct memory line access

Proposal: Flat memory for CXL GPUs

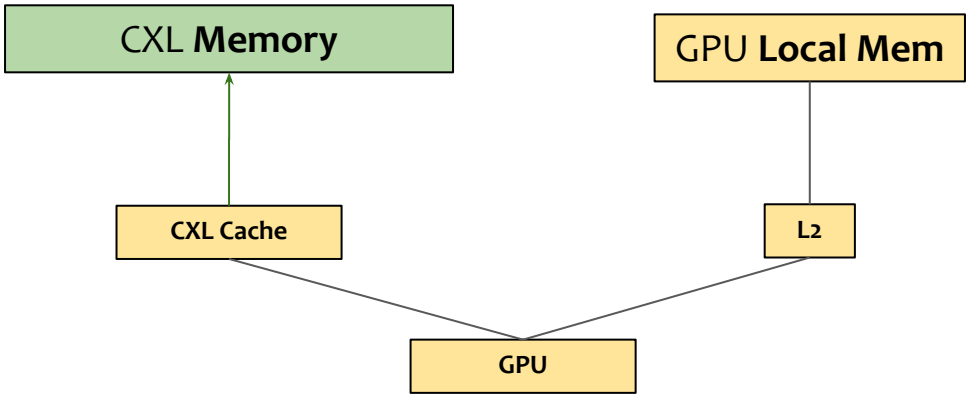
Use GPU local HBM **as coherent cache** for CXL memory

Goals:

1. Programming transparency
2. Efficiency
3. Correctness

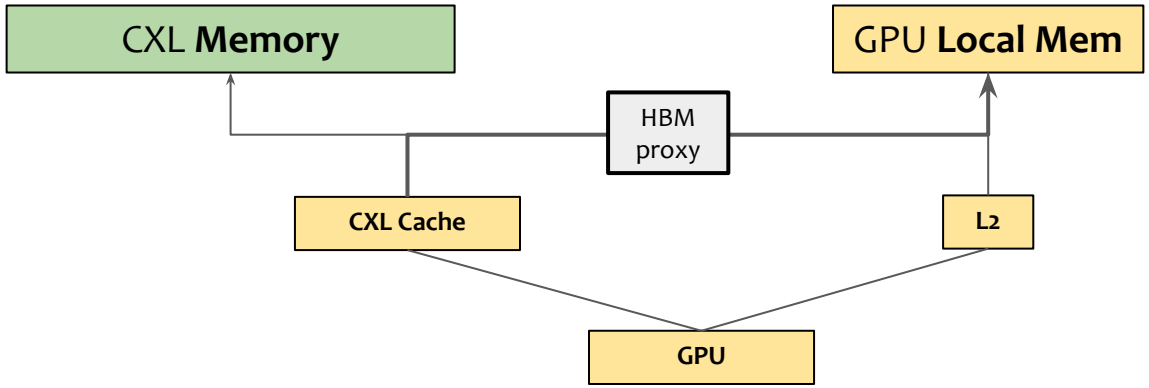
State-of-the-art

CXL and GPU memory are separate domains



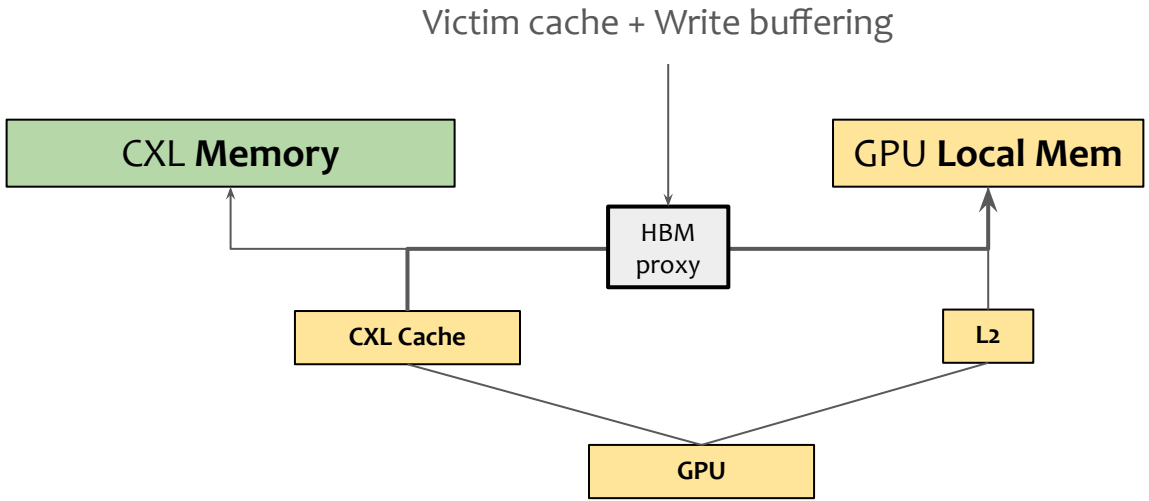
Key idea

Add a module to cache fetched data in the GPU local memory



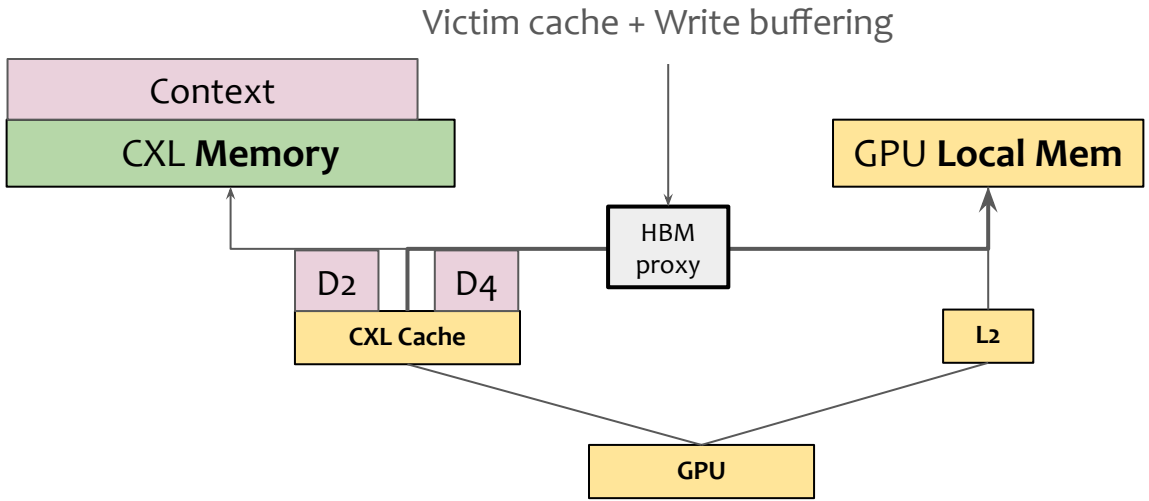
Key Idea

- Let CXL memory cache lines be retained on GPU local HBM
  - **More-reuse** – Larger effective cache capacity
  - **Faster WB/eviction** – CXL memory off the critical path



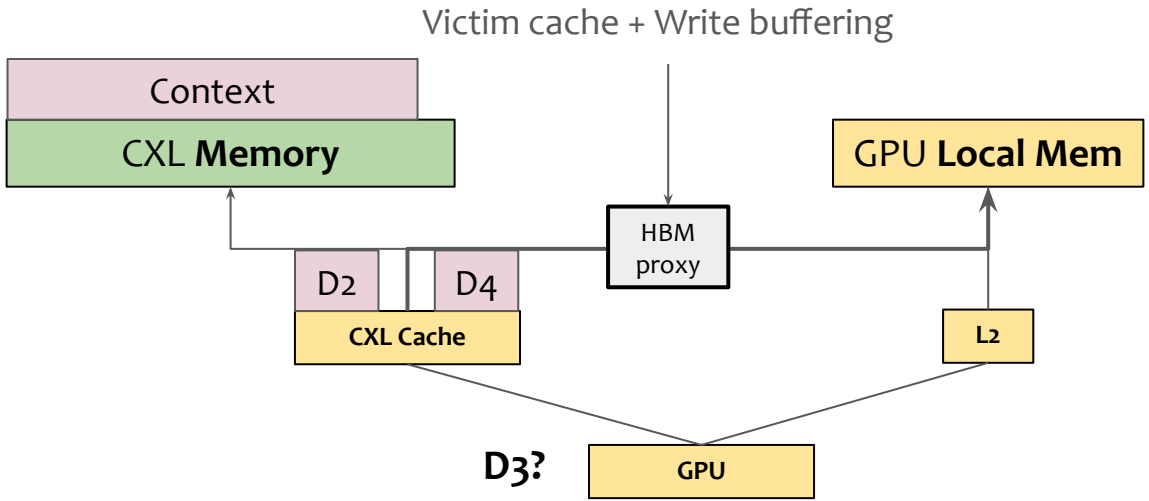
Example: KV-cache offloading

- All context is CXL memory resident
- CXL cache is full with cache lines (D2, D4)



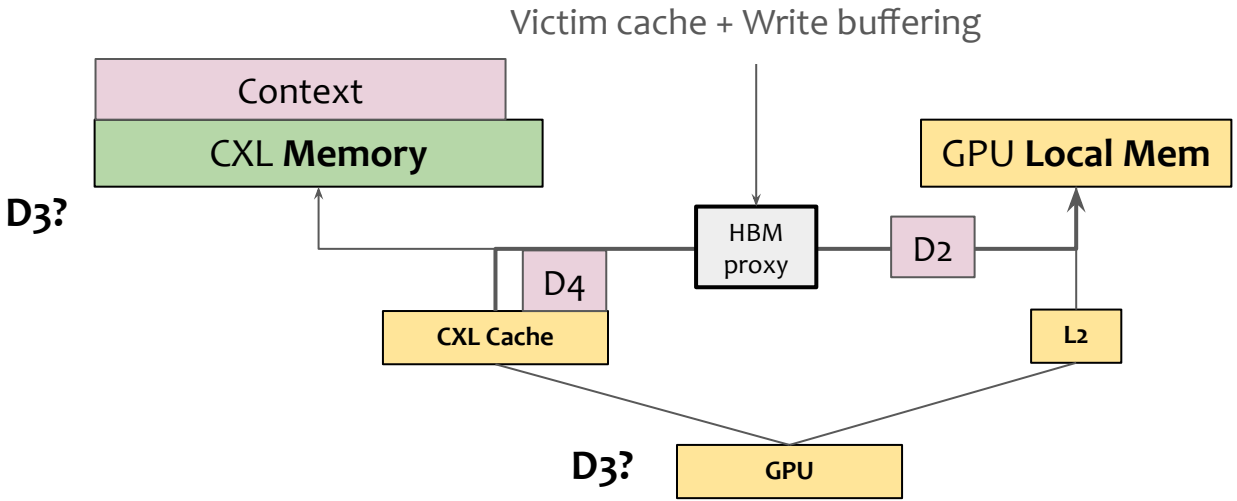
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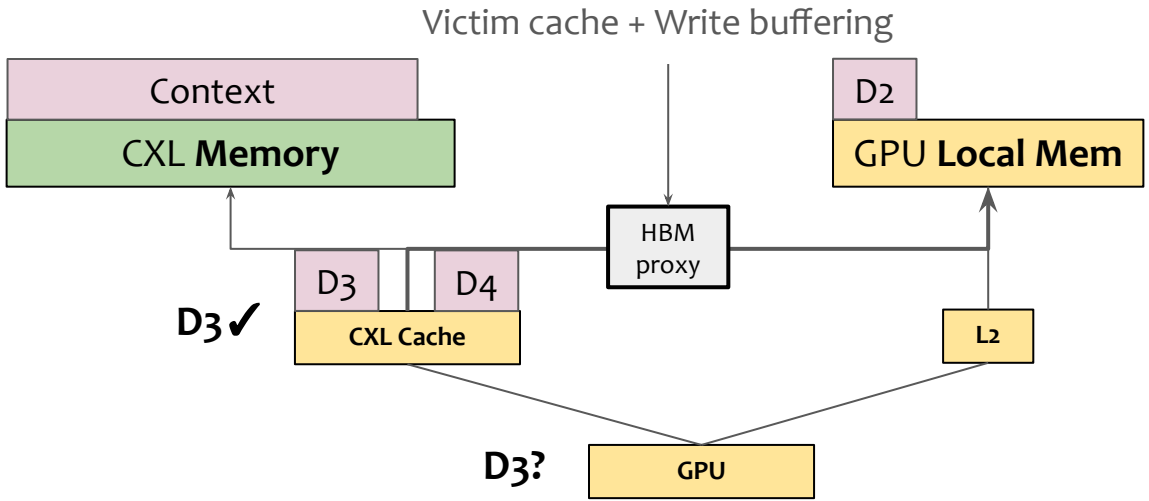
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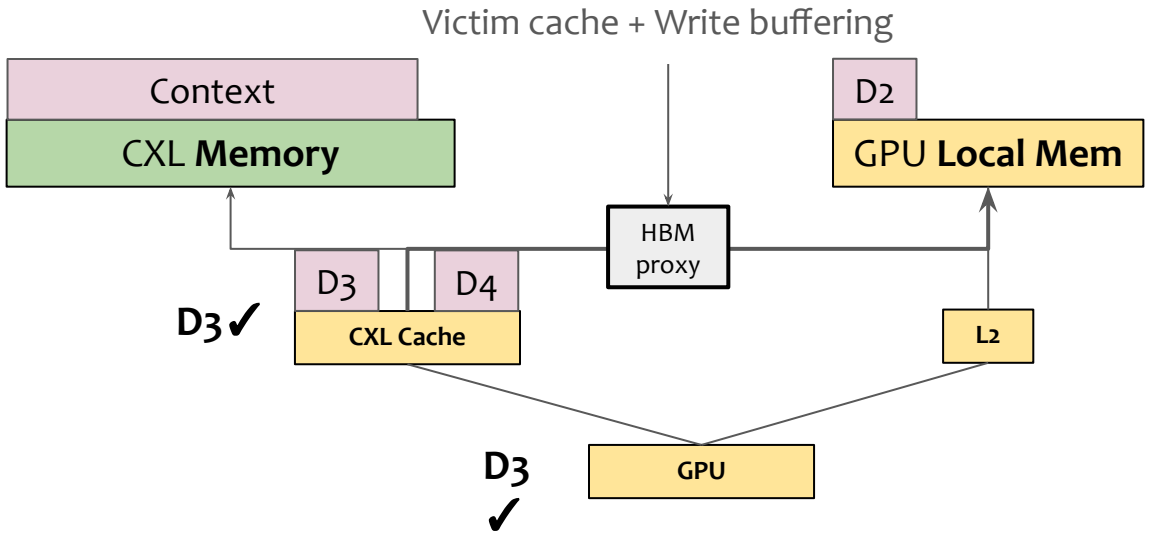
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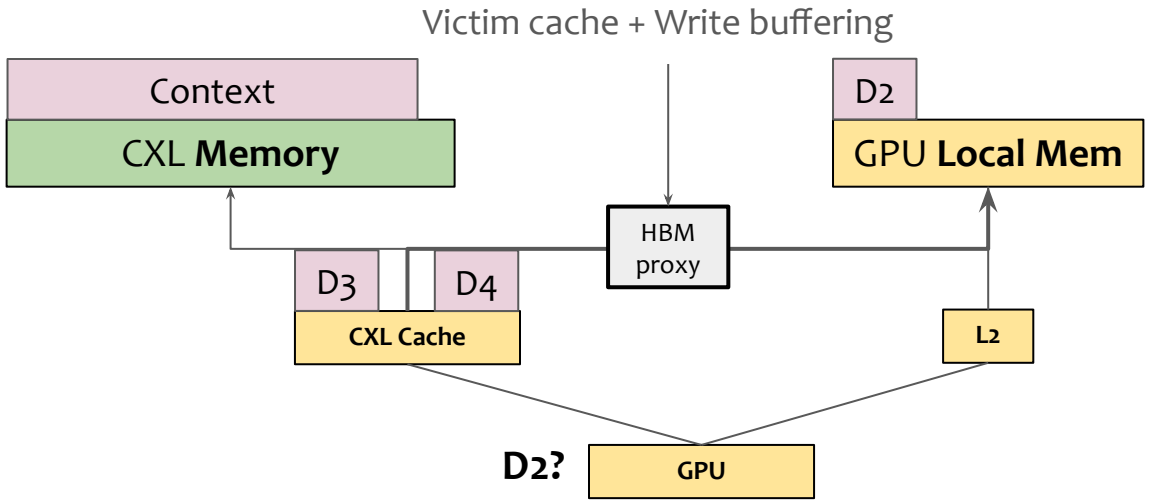
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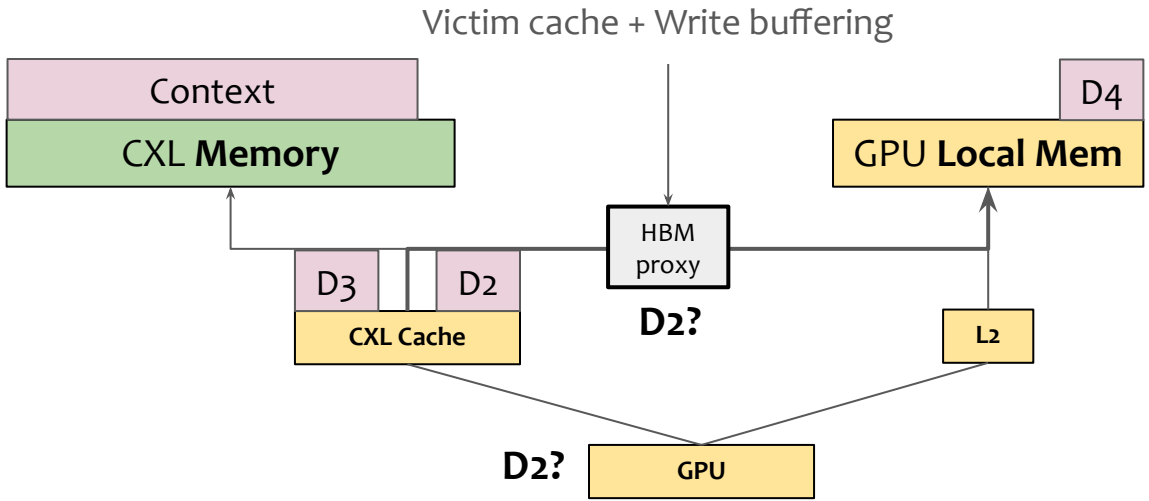
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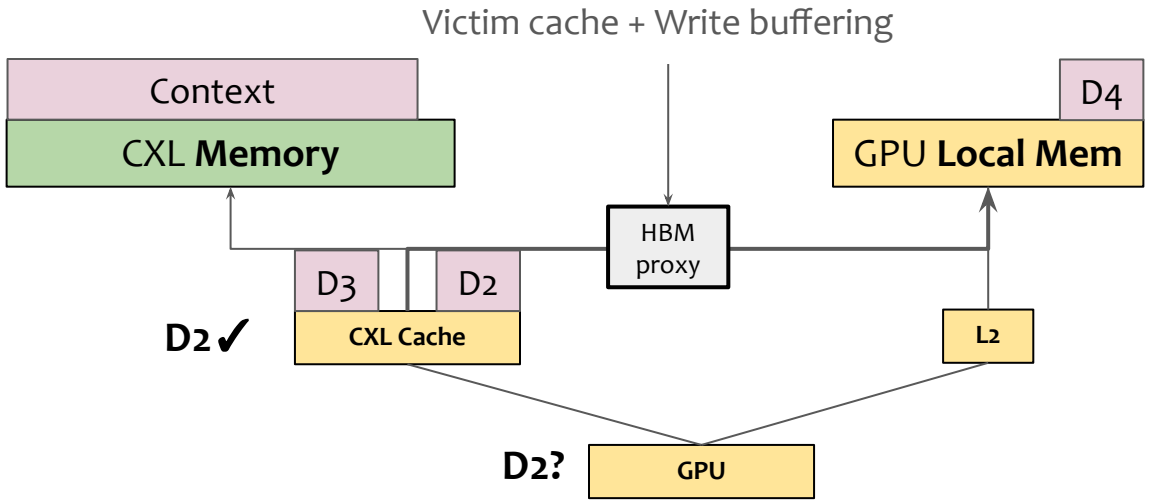
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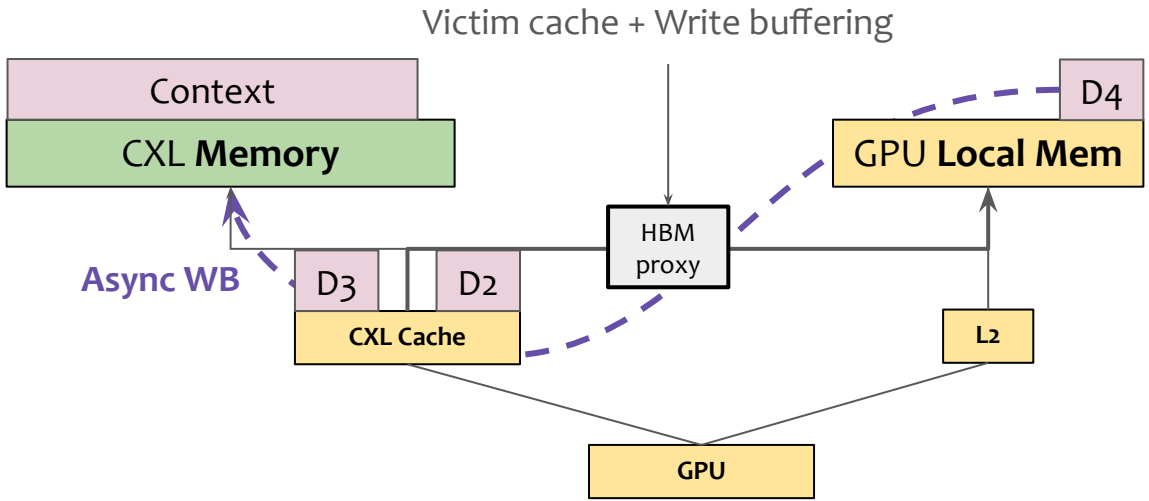
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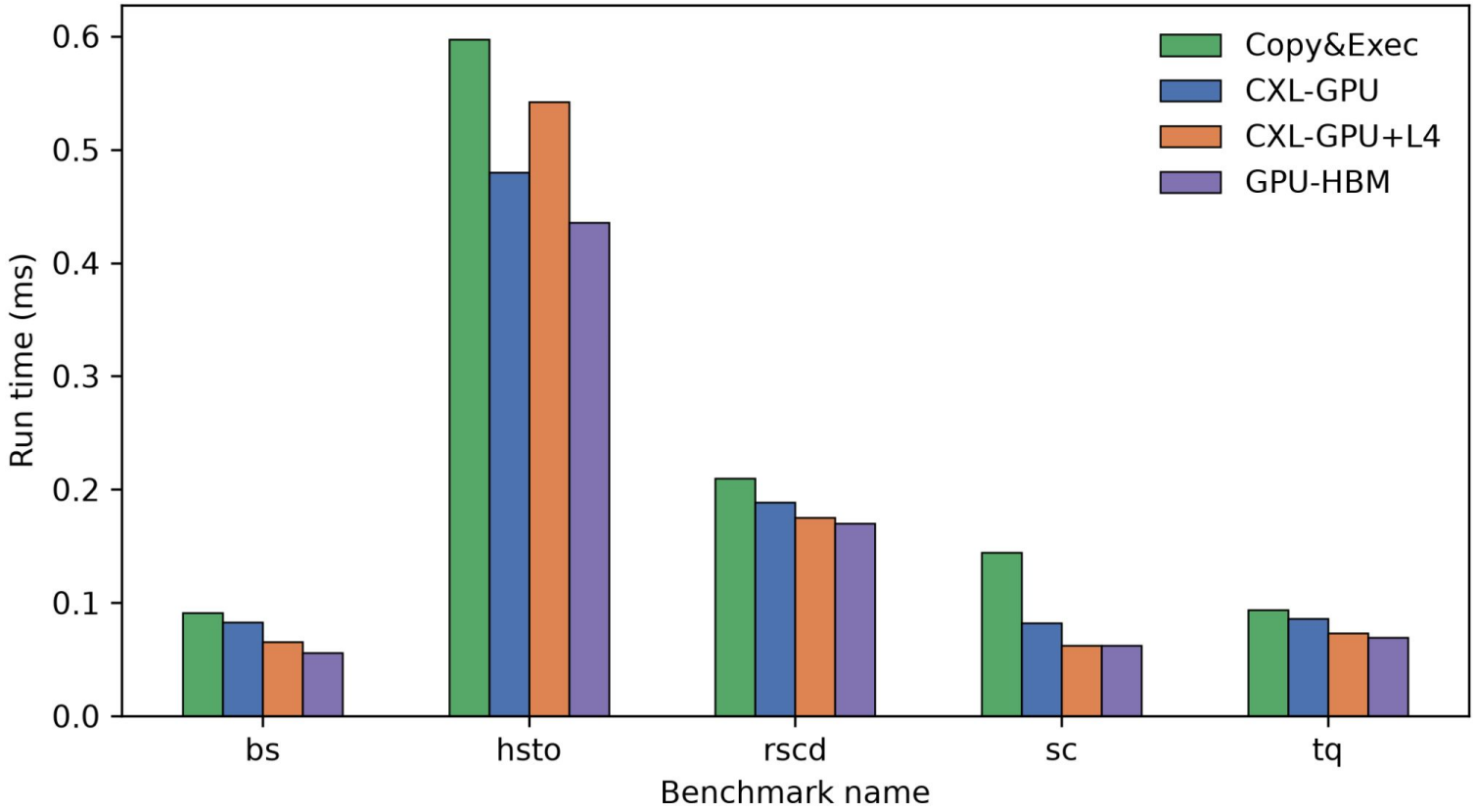


Implementation and Validation

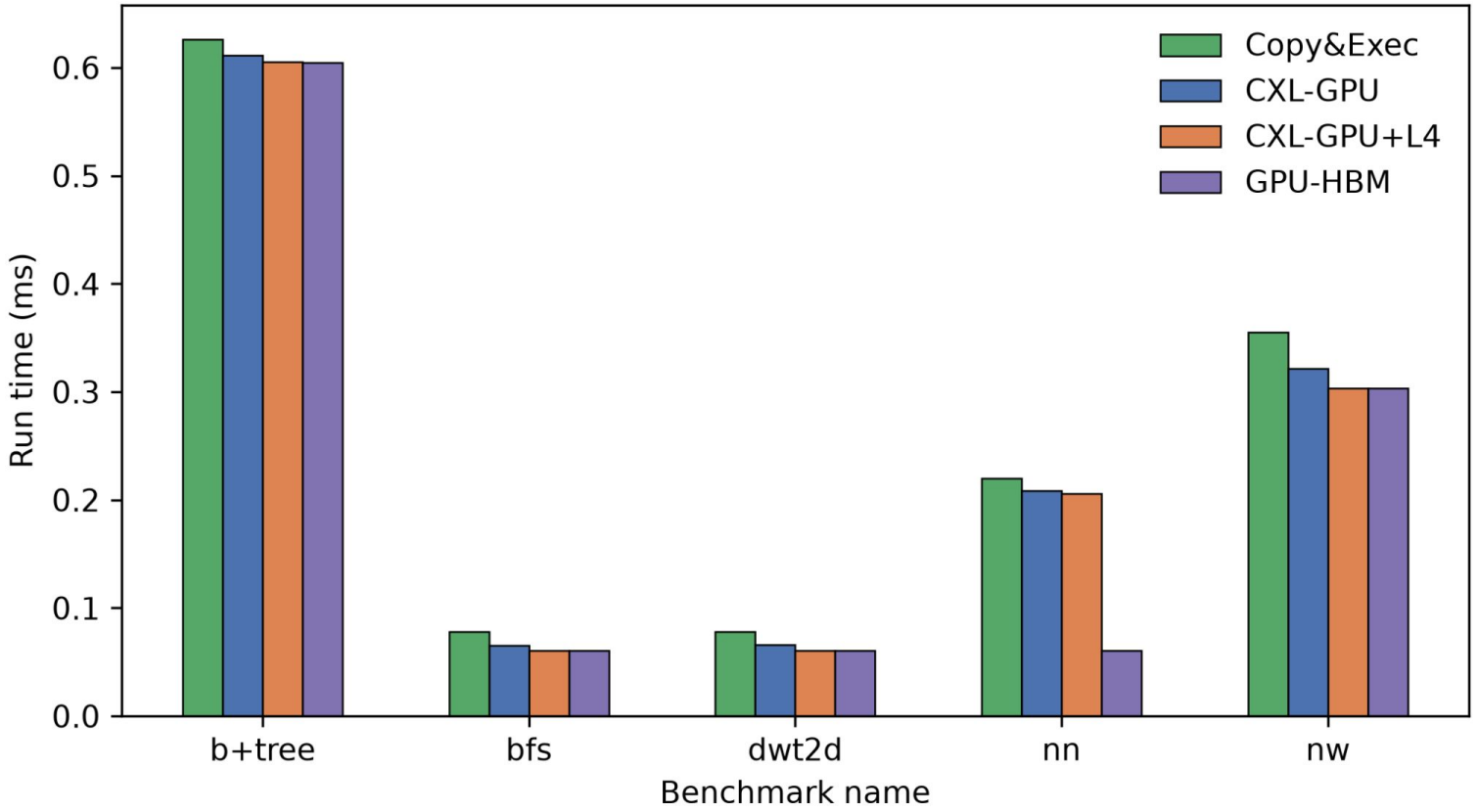
- **Implementation:** coherence protocol extension in gem5 (SLICC)

- **Validation:**
  - CHAI benchmarks (collaborative CPU-GPU kernels)
  - Rodinia benchmarks (scientific computations)

Evaluation: Chai Benchmarks



Evaluation: Rodinia Benchmarks



Conclusion

- CXL-enabled GPUs show better performance compared to the basic system
- CXL memory is a valid option for GPU memory expansion
- Flat memory mode is capable of introducing further optimization to the CXL-enabled GPUs

Try it out!

GitHub